

Task 1

```
zainwright@cvnp1601-lab:~/week04-access-lockdown$ { ls -la /etc/shadow /usr/bin/passwd /tmp /project;
umask; } > permission-audit.txt
cat permission-audit.txt
-rw-r----- 1 root shadow      1153 Sep 21 16:38 /etc/shadow
-rwsr-xr-x  1 root root        64152 May 30  2024 /usr/bin/passwd

/project:
total 8
drwxr-xr-x  2 root developers 4096 Sep 21 16:39 .
drwxr-xr-x 24 root root        4096 Sep 21 16:39 ..

/tmp:
total 52
drwxrwxrwt 13 root root 4096 Sep 21 16:39 .
drwxr-xr-x 24 root root 4096 Sep 21 16:39 ..
drwxrwxrwt  2 root root 4096 Sep 21 16:29 .font-unix
drwxrwxrwt  2 root root 4096 Sep 21 16:29 .ICE-unix
drwx----- 2 root root 4096 Sep 21 16:29 snap-private-tmp
drwx----- 3 root root 4096 Sep 21 16:29 systemd-private-497d8c9e56eb4a2a93389e125ce4040f-ModemManage
r.service-o3xod6
drwx----- 3 root root 4096 Sep 21 16:29 systemd-private-497d8c9e56eb4a2a93389e125ce4040f-polkit.serv
ice-HRW30v
drwx----- 3 root root 4096 Sep 21 16:29 systemd-private-497d8c9e56eb4a2a93389e125ce4040f-systemd-log
ind.service-3Alon5
drwx----- 3 root root 4096 Sep 21 16:29 systemd-private-497d8c9e56eb4a2a93389e125ce4040f-systemd-res
olved.service-d2rRHL
drwx----- 3 root root 4096 Sep 21 16:29 systemd-private-497d8c9e56eb4a2a93389e125ce4040f-systemd-tim
esyncd.service-qVVCfV
drwx----- 2 root root 4096 Sep 21 16:29 vmware-root_750-2957714542
drwxrwxrwt  2 root root 4096 Sep 21 16:29 .X11-unix
drwxrwxrwt  2 root root 4096 Sep 21 16:29 .XIM-unix
0002
zainwright@cvnp1601-lab:~/week04-access-lockdown$ |
```

Item 1.) /etc/shadow

String: -rw-r-----

Sections: (-) Regular file, (rw-) Owner: root) Read/write perms CANNOT execute, (r--) Group: Shadow) CAN read CANNOT write or execute, (---) **Others**) No permissions.

Special Bits: None.

Purpose: The file contains password hashes and can only be modified by root. Only Root and the shadow group are capable of reading it. Nobody else has access to it to prevent credential thefts or unauthorized manipulation.

Item 2.) /usr/bin/passwd

String: -rwsr-xr-x

Sections: (-) Regular file, (rws) Owner: root) read/write/execute perms with a SUID bit set, (r-x) Group: root) read/execute, (r-x) Others) read/execute.

Special Bits: SUID (s)

Purpose: The passwd program needs to be able to modify /etc/shadow, which standard users are unable. SUID allows all the user to update their own password with temp root privileges.

Item 3.) /tmp/project

String: drwxr-xr-x

Sections: (d) directory, (rwx) Owner: root) full control, (r-x) Group: developers) read and execute, (r-x) others) read and execute.

Special Bits: None.

Purpose: The directory is readable to anyone and they are able to execute as well, but it is only writeable by root to prevent unauthorized modification.

Item 4.) /tmp

String: drwxrwxrwt

Sections: (d) directory, (rwx) Owner: root) full control, (rwx) Group: root) full control, (rwx) others) full control excluding sticky bit rules.

Special Bits: Sticky bit (t)

Purpose: /tmp needs to be writeable by anyone. The sticky bit prevents others from tampering with files that are not theirs.

Umask Math: Umask: 0002, New Files 664 (rw-rw-r--), new directories = 775 (rwxrwxr-x)

Task 2

```
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ touch ~/deploy.sh
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ ls -la ~/deploy.sh
-rw-rw-r-- 1 zainwright zainwright 0 Sep 24 04:11 /home/zainwright/deploy.sh
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ chmod 750 ~/deploy.sh
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ ls -la ~/deploy.sh
-rwxr-x--- 1 zainwright zainwright 0 Sep 24 04:11 /home/zainwright/deploy.sh
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ sudo chmod g+w /project
[sudo] password for zainwright:
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ sudo chown $(whoami):developers /project
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$ ls -ld /project
drwxrwxr-x 2 zainwright developers 4096 Sep 24 02:02 /project
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week04-access-lockdown$
```

The octal is made up of 750. 7=Full control (rwx), 5=read/execute (r-x), and 0=no permissions (---) and would be combined into rwxr-x--- in this octal the owner has full control, the group has limited access, and the others are locked out.

Task 3

```
zainwright@cvnp1601-lab:~$ sudo chmod +t /project
zainwright@cvnp1601-lab:~$ su - alice
Password:
$ cd /project
$ nano alice-file.txt
$ su - bob
Password:
$ cd /project
$ ls -la
total 12
drwxrwxr-t 2 zainwright developers 4096 Sep 24 05:34 .
drwxr-xr-x 24 root          root      4096 Sep 24 02:02 ..
-rw-rw-r-- 1 alice         alice      4 Sep 24 05:34 alice-file.txt
$ rm alice-file.txt
rm: remove write-protected regular file 'alice-file.txt'? y
rm: cannot remove 'alice-file.txt': Operation not permitted
$ cd /project
$ ls -ld
drwxrwxr-t 2 zainwright developers 4096 Sep 24 05:34 .
$ ls -la
total 12
drwxrwxr-t 2 zainwright developers 4096 Sep 24 05:34 .
drwxr-xr-x 24 root          root      4096 Sep 24 02:02 ..
-rw-rw-r-- 1 alice         alice      4 Sep 24 05:34 alice-file.txt
$ |
```

Sticky will not let a user tamper with work that is not theirs.

Task 4:

```
$ cd /project
$ nano before-sgid.txt
$ sudo chmod g+s /project
$ nano after-sgid.txt
$ ls -la
total 20
drwxrwsr-t 2 zainwright developers 4096 Sep 24 05:48 .
drwxr-xr-x 24 root          root      4096 Sep 24 02:02 ..
-rw-rw-r-- 1 alice         developers 7 Sep 24 05:48 after-sgid.txt
-rw-rw-r-- 1 alice         alice      4 Sep 24 05:34 alice-file.txt
-rw-rw-r-- 1 alice         developers 7 Sep 24 05:47 before-sgid.txt
$ |
```

(I screwed this one up because I ran the chmod command in the home directory on accident, but we can still see the change with the original .txt we made.) In a nutshell SGID is SUID but for groups instead of users. With this rule any new files stay in the group.

Task 5:

Pushed file to git.